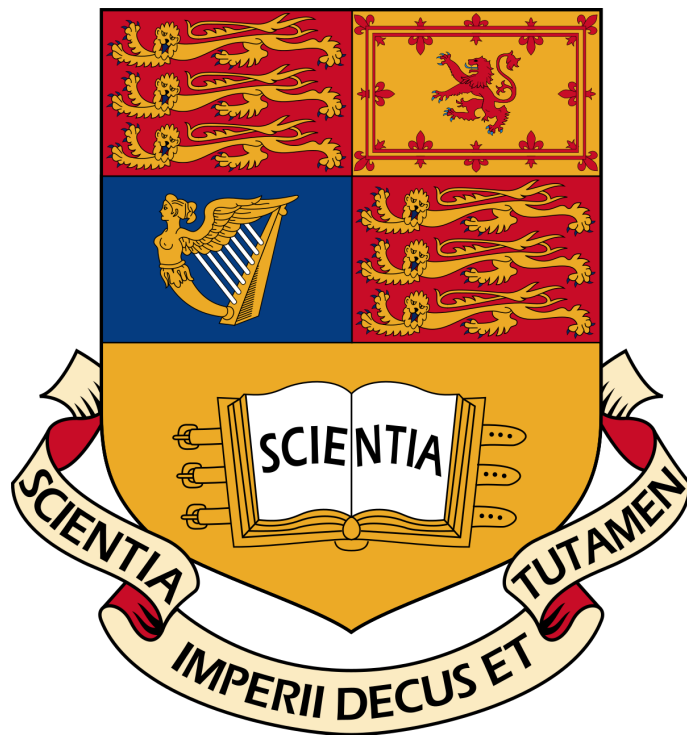


# CANONICAL QUANTISATION OF GRAVITATIONAL SYSTEMS

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## Declaration

The research presented in this thesis is based on work carried out at the Department of Physics, Imperial College London by the author and Prof. João Magueijo. The results of **Chapter 5** will appear in [\[1\]](#).

Ganga Singh Manchanda

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## Abstract

We provide a brief introduction to concepts in Canonical Quantum Gravity including constraints, Dirac quantisation, ADM decomposition, and Superspace. We apply these ideas to FLRW spacetime to describe, in a quantum manner, a homogeneous and isotropic cosmology, specifically a closed universe permeated by a massless scalar field. Through the Einstein-Hilbert action, we produce a gravitational Hamiltonian which upon quantisation leads to a Wheeler-DeWitt equation. Applying the WKB approximation, we solve for the wavefunction and construct Gaussian wavepacket states centred on a classical constant of motion. The ridges of these wavepacket states coincide with the classical configuration space trajectory corresponding to our choice of constant of motion. Through the study of this system we develop machinery and interpretation to inform our investigation of quantum fluctuations of the Schwarzschild metric. We begin with a generic spherically symmetric and static spacetime which is Schwarzschild on-shell. The gravitational Hamiltonian we find must be canonically transformed before it is appropriate to quantise. The new canonical co-ordinates cover a negative radius region of classical Schwarzschild spacetime past asymptotic infinity. We thus interpret asymptotic infinity as a wormhole and explore the region in both the classical and quantum theories. This exploration is a work in progress.

# Contents

|          |                                                     |           |
|----------|-----------------------------------------------------|-----------|
| <b>1</b> | <b>Prologue</b>                                     | <b>2</b>  |
| 1.1      | Motivation . . . . .                                | 2         |
| 1.2      | The First Approach . . . . .                        | 3         |
| <b>2</b> | <b>Constrained Systems</b>                          | <b>4</b>  |
| 2.1      | The Failed Legendre Transform . . . . .             | 4         |
| 2.2      | Gauge Freedom in Constraints . . . . .              | 5         |
| 2.3      | Dirac's Quantisation . . . . .                      | 6         |
| 2.4      | Imposing Symmetries on Metrics . . . . .            | 6         |
| <b>3</b> | <b>Canonical Gravity</b>                            | <b>8</b>  |
| 3.1      | 3 + 1 Decomposition . . . . .                       | 8         |
| 3.2      | The Vanishing Hamiltonian . . . . .                 | 9         |
| 3.3      | Hypersurface Deformation from Constraints . . . . . | 10        |
| 3.4      | Boundary Measures of Mass . . . . .                 | 11        |
| <b>4</b> | <b>Copernican Spacetime</b>                         | <b>12</b> |
| 4.1      | Homogeneity & Isotropy . . . . .                    | 12        |
| 4.2      | The Hilbert Space of a Universe . . . . .           | 12        |
| 4.3      | General Behaviour of Wheeler-DeWitt . . . . .       | 13        |
| 4.4      | Hamiltonian Cosmology . . . . .                     | 14        |
| 4.5      | Minisuperspace Wavepackets . . . . .                | 15        |
| <b>5</b> | <b>Schwarzschild Spacetime</b>                      | <b>18</b> |
| 5.1      | Spherical Symmetry & Staticity . . . . .            | 18        |
| 5.2      | Classical Formulation . . . . .                     | 19        |
| 5.3      | Wormhole Co-ordinates . . . . .                     | 20        |
| 5.4      | Quantisation . . . . .                              | 24        |
| 5.5      | Wavepacket Analysis . . . . .                       | 25        |
| <b>6</b> | <b>Conclusion</b>                                   | <b>27</b> |
| 6.1      | Overview . . . . .                                  | 27        |
| 6.2      | Outlook . . . . .                                   | 28        |
| <b>A</b> | <b>Dead Ends</b>                                    | <b>29</b> |
| A.1      | The Inclusion of Boundary Terms . . . . .           | 29        |
| A.2      | Reduced Phase Space Quantisation . . . . .          | 30        |
| A.3      | Inverse Momentum Quantisation . . . . .             | 31        |

# Chapter 1

## Prologue

### 1.1 Motivation

The early twentieth century saw two unprecedented revolutions in physics to upheave the Newtonian notions of the time. These revolutions, the theories of General Relativity and Quantum Mechanics, have since developed into extraordinary predictors on their respective length scales. They do, however, have their failings and they are fundamentally incompatible. The next revolution in physics, a theory of Quantum Gravity, should break the stalemate between these two, such that the laws which predict phenomena on the Planck scale,  $10^{-35}$  m, could equally be applied on cosmological scales,  $10^{26}$  m.

This nascent revolution faces many obstructions; a direct unification of the two theories is not possible for a number of reasons. By far the greatest division to be resolved centres around time [2]. In General Relativity, time and space are equally malleable and deformable, dynamically unified under the geometry of spacetime. In Quantum Mechanics time is an immutable background parameter forbidden by Pauli's theorem from being anything more [3]. A theory which is both quantum and geometric in nature must reconcile these differences. What could go wrong?

The insufficiencies of General Relativity are laid bare in the study of cosmology and black holes, both of which are plagued by singularities. We naively think of singularities as points where the curvature, in the form of the Kretschmann scalar, diverges. More fundamentally they signal a complete breakdown of the theory in geometries in which a geodesic can simply end (or start), that is, in geodesically incomplete spacetimes [4].

In the context of cosmology, any geodesic can be traced back to the exact same 'big bang' singularity. General Relativity categorically fails to address the origin of this singularity, let alone resolve issues concerning features of the present-day universe, such as the horizon problem in the Cosmic Microwave Background [5]. General Relativity also cannot tell us the fate of an observer heading towards the singularity, a possibility in black hole spacetimes. A geodesic observer that reaches the singularity is obstructed from evolving further within the spacetime, there is none left to evolve into.

All this is to say we need a better solution: A theory of Quantum Gravity. This can be approached through four methods and most commonly, we see the first and third employed [6]. The first is to treat geometry as fundamental and apply the quantisation schemes set out [7, 8] to General Relativity as with any classical theory. The third is to treat Quantum Field Theory as fundamental and build a theory in which General Relativity is recovered in the low energy

limit. Both approaches must account for the novel features of each theory and enhance them to answer the questions that neither can [9, 10].

## 1.2 The First Approach

In this thesis, we follow the first approach, studying the quantisation of gravitational systems. We look specifically at canonical rather than covariant quantisation, which imposes a  $3 + 1$  split of space and time.

Historically, the first approach has seen great success in cosmology. Through quantisation the cosmological singularity can be bypassed in a number of ways, including the creation of the universe via tunnelling [11], superpositions of expanding and contracting universes [12, 13], and models of ‘bouncing’ universes [14]. Some recent approaches bypass the singularity using scalar fields. Classically, the ad-hoc inclusion of scalar fields in cosmology has alleviated certain issues, as with the inflaton field and the horizon problem [15]. Quantum mechanically the possibilities are much richer, with the scalar field able to act as a physical clock, an intrinsic measure of time, similar to the unimodular gravity approach [16]. Measuring time with respect to the scalar field clock, the singularity can be resolved [17].

These quantum methods have been widely explored in cosmology but less so in the context of black holes. Thus far, the treatment of Quantum Black Holes has generally followed a semi-classical approach wherein Quantum Field Theory is performed on a fixed Schwarzschild background. This scheme, as a first order approximation of Quantum Gravity, has been surprisingly successful, predicting Hawking radiation [18] and the Bekenstein-Hawking entropy [19]. We develop the other half of this approximation. We quantise the background, treating the metric components as canonical variables that can stray from the classical Schwarzschild solution off-shell. Such an analysis has been performed with differing restrictions in Superspace using path integrals [20, 21], canonical methods [22, 23, 24, 25], and the loop representation [26, 27]. These methods cover topics such as singularity resolution, horizon fluctuations, and black hole thermodynamics.

The structure of the thesis is as follows. In [Chapter 2](#) we summarise the results of [7, 28] on the study of constrained systems, where the Hamiltonian derived through the Legendre transform cannot completely capture the physics. We review how gauge systems are constrained and the implications of gauge fixing [29]. In [Chapter 3](#) we derive the Hamiltonian formulation of General Relativity, imposing a  $3 + 1$  canonical topology on the spacetime and following its implications through the Einstein-Hilbert action to a gravitational Hamiltonian [30, 31]. We find that, in general, the Hamiltonian vanishes such that it cannot generate time translation, adding to the ambiguity surrounding time in Quantum Gravity. In [Chapter 4](#) we review the canonical quantisation of homogeneous and isotropic cosmologies by applying the results derived to FLRW spacetime [32]. Rather than cover boundary proposals like many reviews of Quantum Cosmology, we look at a specific wavepacket formulation by [33] which will inform our later work. Finally, in [Chapter 5](#), inspired by methods in Quantum Cosmology, we perform a canonical quantisation of spherically symmetric spacetime similar to [22, 23, 24, 25] who all develop theories of Quantum Black Holes. Rather than look at the black hole itself, we investigate asymptotic infinity and find a wormhole. This wormhole is analysed from both the classical and quantum viewpoint. This analysis is an ongoing work in progress.

This thesis will largely follow the sign conventions of [34].

# Chapter 2

## Constrained Systems

### 2.1 The Failed Legendre Transform

In many systems, the Hamiltonian,  $\mathcal{H}(q, p, t)$ , derived through a Legendre transform of the Lagrangian,  $\mathcal{L}(q, \dot{q}, t)$ , is not a complete statement of the theory. Consider the conjugate momenta,

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}^i}. \quad (2.1)$$

When the conjugate momenta cannot be inverted to find the generalised velocities of coordinates, there must exist relations of the form  $\phi(q, p) = 0$  (implied by the momenta expressions) which cannot be inferred from the Hamiltonian but nevertheless express real physics. These relations, ‘primary constraints’, are satisfied only on a subspace of the phase space called the constraint surface [28]. This is signified by the weak equality,  $\phi(p, q) \approx 0$ . We call a trajectory on the constraint surface ‘on-shell’, corresponding to a classical solution of the system. Quantum behaviour that deviates from the classical solution is found off the constraint surface or ‘off-shell’. The existence of primary constraints is confirmed by a degenerate Hessian,

$$\det \frac{\partial p_i}{\partial \dot{q}_j} = 0. \quad (2.2)$$

Intuitively, we understand that if this determinant vanishes then we cannot directly map generalised velocities and conjugate momenta, and thus cannot pass between the Lagrangian and Hamiltonian.

We treat primary constraints as canonical variables and add them to the Hamiltonian using Lagrange multipliers. As they vanish classically, the new Hamiltonian constitutes a physically identical theory, but we have, by hand, inserted physically important dynamics not captured by the Legendre transform. The total Hamiltonian is given by the sum of the naive Hamiltonian and the primary constraints [7]

$$\mathcal{H}_T = \mathcal{H} + c_i \phi_i. \quad (2.3)$$

Time derivatives are handled with the usual Poisson bracket and this total Hamiltonian. We expect that the constraints vanish at all times such that we impose the consistency condition

$$\dot{\phi}_i = \{\phi_i, H_T\} \approx 0. \quad (2.4)$$

Enforcing this condition may generate another constraint, a ‘secondary constraint’, or it may fix one of the Lagrange multipliers. Whether primary or secondary we classify the ‘first-class’

subset of constraints, more generally functions  $F(q, p)$ , as those whose subalgebra closes:

$$\{\phi_i, \phi_j\} = f_{ij}^k \phi_k. \quad (2.5)$$

All other constraints are ‘second-class’. If the secondary constraint generated is first-class, we add it to the total Hamiltonian and check once again for the constancy of all constraints in a recursive program that continues until the consistency checks have produced all possible constraints and instead determine Lagrange multipliers. Once the program terminates, we are left with an extended Hamiltonian,  $\mathcal{H}_E$ , made up of the naive Hamiltonian and all first-class constraints, primary and secondary, with both arbitrary and fixed Lagrange multipliers. This Hamiltonian captures all physical degrees of freedom of the system.

## 2.2 Gauge Freedom in Constraints

Constraints are an expression of dependence between the conjugate momenta and generalised velocities where we expect none. In this sense, acknowledging the constraints undoes an over-counting of the physical degrees of freedom in the system.

We find that first-class constraints, specifically constraints whose Lagrange multipliers are left undetermined, generate symmetries of the system. This is true because although a given  $(q, p)$  must specify a state, a given state can correspond to any  $(q, p)$  as long as the arbitrary functions,  $c_i$ , can absorb the physical change of state. These are gauge transformations; they are unobservable in the physical state. On the other hand, quantities  $F(q, p)$  that do not change under these transformations are observables simply because they cannot be gauge degrees of freedom [34]:

$$\{F, \phi_i\} \approx 0 \quad \forall i. \quad (2.6)$$

Gauge degrees of freedom are mathematically described by constraints.

The objective of gauge fixing is to remove ‘unobservable’ degrees of freedom and reduce the phase space such that there is a one-to-one correspondence between a given  $(q, p)$  and a physical state. This is done by imposing gauge conditions, in the form of constraints, by hand. To fully fix the gauge, we need to impose as many gauge conditions as there are first-class gauge degrees of freedom. In doing so, we eliminate all first-class constraints from the system and leave only the new constraints. The remaining constraints must necessarily be second-class as the presence of first-class constraints would imply remaining gauge freedom. Thus, in general, gauge fixing turns first-class constraints into second-class constraints.

As noble a cause as gauge fixing is, reducing the system to its true degrees of freedom and simplifying the physics, the act can be detrimental. Gauge conditions can be imposed on equations of motion after the action has been varied, a ‘post-fixing’ of the gauge, but if finding the equations of motion is difficult, then it is ideal to fix the gauge before varying, ‘pre-fixing’. The subtlety is that a pre-fixing may either be complete or incomplete [29]. An incomplete gauge fixing may inadvertently eliminate equations of motion and generate erroneous degrees of freedom, a disastrous result. A complete gauge fixing avoids this but regardless, generates second-class constraints which introduce difficulty during the quantisation process. It is also non-trivial to determine whether a gauge fixing is complete or incomplete. In the study of gravitational systems, selecting a non-generic metric ansatz constitutes a pre-fixing of the gauge. We avoid gauge fixing where possible to prevent added difficulties.

## 2.3 Dirac's Quantisation

In an unconstrained system, the process of quantisation involves translating canonical variables into operators, Poisson brackets into commutation relations, the Hamiltonian into a Schrödinger equation, and introducing a state,  $|\psi\rangle \in \mathbb{H}$ , that exists in a Hilbert space, to replace the classical idea of trajectories in phase space. In short,

$$q^i, p_i \rightarrow \hat{q}^i, \hat{p}_i, \quad (2.7)$$

$$\{q^i, p_j\} = \delta_j^i \rightarrow [\hat{q}^i, \hat{p}_j] = i\hbar\delta_j^i, \quad (2.8)$$

$$\hat{\mathcal{H}}|\psi\rangle = i\hbar|\dot{\psi}\rangle. \quad (2.9)$$

All operators in the quantum theory act on the state,  $|\psi\rangle$  and obey the commutation relations. The presence of constraints adds some subtlety to this process. In the presence of second-class constraints, not all non-vanishing Poisson brackets should correspond to non-vanishing commutators. The original Poisson bracket structure necessarily includes the dependent degrees of freedom that the constraints are intended to eliminate. We thus define the Dirac bracket which projects out 'false' degrees of freedom from the Poisson bracket by projecting onto the true constraint surface,

$$C^{\alpha\beta} = C_{\alpha\beta}^{-1} = (\{\phi_\alpha, \phi_\beta\})^{-1}, \quad (2.10)$$

$$\{F, G\}_D = \{F, G\} - \{F, \chi_\alpha\}C^{\alpha\beta}\{\chi_\beta, G\}, \quad (2.11)$$

where  $\chi_\alpha, \chi_\beta, \dots$  are the subset of second-class constraints of  $\phi_\alpha$ , the set of all constraints [28].

Quantisation of a constrained system is performed in analogy with the earlier scheme but the primary constraints are translated to operators along with the traditional canonical variables, the Dirac bracket replaces the Poisson bracket and the extended Hamiltonian replaces the naive Hamiltonian. That is,

$$q^i, p_i, \phi_i \rightarrow \hat{q}^i, \hat{p}_i, \hat{\phi}_i, \quad (2.12)$$

$$\{q^i, p_j\}_D = \delta_j^i \rightarrow [\hat{q}^i, \hat{p}_j]_D = i\hbar\delta_j^i, \quad (2.13)$$

$$\hat{\mathcal{H}}_E|\psi\rangle = i\hbar|\dot{\psi}\rangle. \quad (2.14)$$

The primary constraints are imposed operator equations once all Dirac brackets are determined. These operator equations restrict the space of physically allowed states. There remain ambiguities in any quantisation procedure due to operator ordering and issues of time, but these are handled on a case-by-case basis.

## 2.4 Imposing Symmetries on Metrics

The formal use of Differential Geometry in General Relativity ensures that we can describe a spacetime independent of a co-ordinate basis, as the pair  $(\mathcal{M}, g)$  [4]. This is useless however when describing systems that exist with respect to co-ordinates,  $\{x^0, x^1, x^2, x^3\}$ . To choose a co-ordinate system is to split the spacetime into hypersurfaces of constant co-ordinates. This split is called a foliation.

The metric of a generic spacetime,  $g_{\mu\nu}$ , consists of ten free functions such that the Einstein-Hilbert action and any gravitational Lagrangian derived from it should possess ten physical degrees of freedom. When we perform a foliation we re-group those ten degrees of freedom

into a format more fitting to the context. To deal with spacetime symmetry, we first perform an appropriate foliation into hypersurfaces of constant symmetric co-ordinate,  $x^i$ . Out of the ten degrees of freedom some are redundant in a symmetric system, effectively becoming gauge degrees of freedom. To impose the symmetry is to eliminate the physical degrees of freedom corresponding to non-symmetric transformations while carefully leaving gauge degrees of freedom unfixed (to avoid second-class constraints). It is not obvious which degrees of freedom are physical and which are gauge. Throughout this thesis we deal with symmetric systems in this manner.

# Chapter 3

## Canonical Gravity

### 3.1 3 + 1 Decomposition

Following the work of [30] we perform a 3 + 1 (ADM) decomposition of Lorentzian spacetime into spacelike Cauchy hypersurfaces each labelled by a ‘time’ co-ordinate imposing the topology

$$\mathcal{M} \cong \mathbb{R}(t) \times \Sigma_t(x^i), \quad (3.1)$$

$$x^\mu = (t, x^i). \quad (3.2)$$

Such a foliation must produce a globally hyperbolic spacetime, ensuring a sufficient domain of dependence [35]. Points on adjacent spacelike hypersurfaces are connected by a timelike vector,  $\delta x^\mu = \dot{x}^\mu \delta t$ , which we decompose into its tangent,  $\delta x^\mu_{\parallel} = N^\mu \delta t$  and normal,  $\delta x^\mu_{\perp} = N n^\mu \delta t$  components as shown in [Figure 3.1](#). We call  $N^\mu$  the shift and  $N$  the lapse function. The lapse,  $N$  scales the unit normal to the hypersurface  $n^\mu$ .

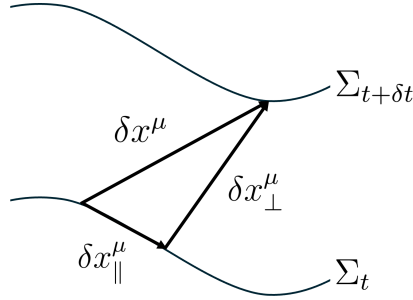


Figure 3.1: Each  $\Sigma_t$  describes a ‘snapshot’ of three-space at a fixed time. The lapse dictates how fast co-ordinate time,  $t$ , runs relative to proper time. The shift determines how points are mapped ‘upwards’ onto the next hypersurface. For a vanishing shift, points are mapped directly ‘upwards’.

The metric induced on  $\Sigma_t$  by  $g_{\mu\nu}$  is called the first fundamental form:

$$h_{\mu\nu} \equiv g_{\mu\nu} + n_\mu n_\nu. \quad (3.3)$$

We notice that between points on adjacent hypersurfaces the triplet  $\{N, N^\mu, h_{\mu\nu}\}$  completely determines the spacetime metric

$$\begin{aligned} ds^2 &= -N^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt) \\ &= -(N^2 - h_{ij}N^i N^j)dt^2 + 2h_{ij}N^j dx^i dt + h_{ij}dx^i dx^j. \end{aligned} \quad (3.4)$$

This is the  $3 + 1$  decomposition of spacetime, a foliation which preserves all ten degrees of freedom we expect. Although we select a ‘time’ direction, it is an arbitrary co-ordinate such that diffeomorphism invariance remains unbroken until we think about physical (proper) time.

The extrinsic curvature of  $\Sigma_t$  embedded in  $\mathcal{M}$  is called the second fundamental form and can be defined through the four-covariant derivative or the Lie derivative

$$K_{\mu\nu} \equiv h_{\mu}^{\rho} \nabla_{\rho} n_{\nu} = \frac{1}{2} \mathcal{L}_{\mathbf{n}} h_{\mu\nu}. \quad (3.5)$$

$K_{\mu\nu}$  is the natural choice for the generalised velocity of  $h_{\mu\nu}$  and this is made apparent on spacelike hypersurfaces where we introduce the three-covariant derivative  $D_i$ ,

$$K_{ij} = \frac{1}{2N} (\dot{h}_{ij} - 2D_{(i} N_{j)}). \quad (3.6)$$

The Gauss-Codazzi relations allow us to write the four-Ricci scalar,  $\mathbf{R}$ , in terms of the second fundamental form and the three-Ricci scalar,  $R$ , of the spacelike hypersurfaces

$$\mathbf{R} = R + K^2 - K^{ij} K_{ij} + (\dots), \quad (3.7)$$

$$(\dots) = 2(\nabla_{\mu} n^{\nu})(\nabla_{\nu} n^{\mu}) - 2\nabla_{\mu}(n^{\nu} \nabla_{\nu} n^{\mu}) - 2(\nabla_{\mu} n^{\mu})(\nabla_{\nu} n^{\nu}) + 2\nabla_{\nu}(n^{\nu} \nabla_{\mu} n^{\mu}), \quad (3.8)$$

where  $K = K_i^i$  [34]. We now take  $\Sigma_t$  to be compact without boundary and will treat the more general case at the end of this chapter. Re-expressing (3.8) in  $K, K_{ij}$ ,

$$(\dots) = 2K^{ij} K_{ij} - 2K^2 - 2\nabla_{\mu}(n^{\nu} \nabla_{\nu} n^{\mu}) + 2\nabla_{\nu}(n^{\nu} K), \quad (3.9)$$

and noting the geodesic nature of  $n^{\mu}$ , we produce a  $3 + 1$  gravitational Lagrangian (density) by plugging into the Einstein-Hilbert action with cosmological constant,  $\Lambda$ ,

$$S_{EH} = \int_{\mathcal{M}} d^4x \sqrt{-g} (\mathbf{R} - 2\Lambda) - 2 \int_{\partial\mathcal{M}} d^3x \sqrt{h} K, \quad (3.10)$$

$$S_{3+1} = \int_{\mathbb{R}} dt \int_{\Sigma_t} d^3x N \sqrt{h} (R + K^{ij} K_{ij} - K^2 - 2\Lambda). \quad (3.11)$$

Thus we have built a canonical action, one which treats time and space unequally. Such a split is necessary as the conjugate momenta, and by extension the Legendre transform, uniquely select a ‘time’ co-ordinate.

## 3.2 The Vanishing Hamiltonian

The gravitational Lagrangian in the action (3.11) can be rewritten in terms of a function of spatial metrics called the DeWitt metric. The DeWitt metric acts on the configuration space of the system rather than the spacetime manifold [31]:

$$G_{ijkl} = \frac{1}{2\sqrt{h}} (h_{ik} h_{jl} + h_{il} h_{jk} - h_{ij} h_{kl}), \quad (3.12)$$

$$G^{ijkl} K_{ij} K_{kl} = \sqrt{h} (K^{ij} K_{ij} - K^2). \quad (3.13)$$

In using the DeWitt metric we put the Lagrangian in a familiar  $\mathcal{L} = T - V$  form, clearly indicating the generalised velocities (the rate at which the three-geometry changes),

$$\mathcal{L} = N [G^{ijkl} K_{ij} K_{kl} + \sqrt{h} (R - 2\Lambda)]. \quad (3.14)$$

Our system is implicitly described by canonical variables  $q^i \in \{h_{ij}, N, N_i\}$ . Using the functional analog of (2.1), we define our conjugate momenta  $p_i \in \{\pi^{ij}, \pi^0, \pi^i\}$ ,

$$\pi^{ij} = G^{ijkl} K_{kl} = \sqrt{h}(K^{ij} - Kh^{ij}), \quad (3.15)$$

$$\pi^0 = 0, \quad (3.16)$$

$$\pi^i = 0. \quad (3.17)$$

The lapse and shift momenta, (3.16) and (3.17), form the primary constraints of the system,  $\phi_0 = \pi^0 \approx 0$  and  $\phi_i = \pi^i \approx 0$ . Performing a Legendre transform, we obtain a naive Hamiltonian (density),

$$\begin{aligned} \mathcal{H} &= \pi^{ij} \dot{h}_{ij} - \mathcal{L} \\ &= N[G_{ijkl}\pi^{ij}\pi^{kl} - \sqrt{h}(R - 2\Lambda)] + N_i[-2D_j\pi^{ij}] \\ &= N\mathcal{H} + N_i\mathcal{H}^i. \end{aligned} \quad (3.18)$$

From the Lagrangian perspective, the lapse and shift multipliers constrain the two quantities,  $\mathcal{H}$  and  $\mathcal{H}^i$ , to vanish. These secondary constraints are called the Hamiltonian and Momentum constraints respectively,

$$\mathcal{H} = G_{ijkl}\pi^{ij}\pi^{kl} - \sqrt{h}(R - 2\Lambda) \approx 0, \quad (3.19)$$

$$\mathcal{H}^i = -2D_j\pi^{ij} \approx 0. \quad (3.20)$$

The extended Hamiltonian is a sum of the primary and secondary constraints, all of which vanish on-shell such that the extended Hamiltonian vanishes on-shell. Herein lies the problem. A vanishing Hamiltonian can never produce a Schrödinger equation upon quantisation and generate time translation. Instead it generates a Wheeler-DeWitt equation,  $\hat{\mathcal{H}}|\psi\rangle = 0$ , such that any solution must necessarily be static. Thus it seems that time has no place in a theory of Quantum Gravity, except perhaps as an intrinsic measure [36] or an emergent property [37].

### 3.3 Hypersurface Deformation from Constraints

Through the analysis of our gravitational Hamiltonian, we have produced four primary constraints and four secondary constraints. The primary constraints,  $\phi$  and  $\phi_i$ , generate translations in  $N$  and  $N_i$ . The secondary constraints,  $\mathcal{H}$  and  $\mathcal{H}_i$ , are more interesting.

The purpose of the Hamiltonian and Momentum constraints are two-fold [38]. First, they reduce the phase space of the system as all constraints do. Second, they determine the evolution of this reduced phase space through their Poisson brackets. Conventionally, the Hamiltonian generates time translations and the momentum generates spatial translations. Consequently, we should expect the Hamiltonian constraint to tell us about the evolution between adjacent hypersurfaces and the Momentum constraints to tell us about evolution within a given spatial hypersurface. This is exactly what we find when we analyse the algebra of constraints [34]:

$$\{\mathcal{H}(x), \mathcal{H}(y)\} = \delta_{,i}(x, y)[h^{ij}(x)\mathcal{H}_j(x) + h^{ij}(y)\mathcal{H}_j(y)], \quad (3.21)$$

$$\{\mathcal{H}_i(x), \mathcal{H}(y)\} = \mathcal{H}(x)\delta_{,i}(x, y), \quad (3.22)$$

$$\{\mathcal{H}_i(x), \mathcal{H}_j(y)\} = \mathcal{H}_j(x)\delta_{,i}(x, y) + \mathcal{H}_i(y)\delta_{,j}(x, y). \quad (3.23)$$

The algebra is closed such that the constraints are all first-class and they correspond to gauge degrees of freedom. It is isomorphic to the deformation algebra of an embedded hypersurface

such that we can assign geometric interpretation to each constraint. The Hamiltonian constraint generates deformations normal to the spatial hypersurface,  $\Sigma_t$ , such that it evolves the hypersurface in time. The Momentum constraints generate infinitesimal deformations tangent to the spatial hypersurface,  $\Sigma_t$ , such that they enforce diffeomorphism invariance within each hypersurface [39].

### 3.4 Boundary Measures of Mass

We now return to the issue of boundary terms, which we so carelessly discarded earlier by eliminating the spatial boundaries of  $\mathcal{M}$ . In a spacetime with temporal boundaries only, we found that the surface term obtained from the total derivative in (3.8) exactly cancels the Gibbons-Hawking-York term, a result with dire consequences. For these compact boundaryless spacetimes, there is no energy whatsoever and the Hamiltonian vanishes. It seems then that any real physics must lie in the boundary, a concept we are familiar with from holography [40].

We consider the possibility that  $\Sigma_t$  is asymptotically flat such that the metric falls off at  $r \rightarrow \infty$  as  $h_{ij} = \delta_{ij} + \mathcal{O}(r^{-1})$ . Under these conditions, there is no longer an exact cancellation of boundary terms. The integrals over the temporal boundaries cancel as before, but the integrals over the spatial boundaries, which were not present earlier, do not cancel. Instead they combine to produce a single boundary term evaluated at spatial infinity which we write as [30, 41]

$$S_{\partial\mathcal{M}} = - \lim_{r \rightarrow \infty} \oint_{S_\infty^2} d^2\sigma^i N(h_{ij,j} - h_{jj,i}). \quad (3.24)$$

This is the fixed boundary term of the action for asymptotically flat spacetimes. Translating this term into the Hamiltonian formalism we find that this boundary term contributes energy to the system. In fact it contributes all the energy to the system as the bulk Hamiltonian is constrained to vanish,

$$\mathcal{H}_\infty = \mathcal{H} + E_{\text{ADM}}, \quad (3.25)$$

$$E_{\text{ADM}} = \lim_{r \rightarrow \infty} \oint_{S_\infty^2} d^2\sigma^i (h_{ij,j} - h_{jj,i}). \quad (3.26)$$

This ‘ADM’ energy encodes the entire mass and gravitational energy content of the system as seen by an observer looking in from spatial infinity. This must be a conserved quantity as no outgoing energy should be able to escape the observer; to do so would require radiation past infinity.

We may also evaluate the energy as seen by an observer at some finite  $r$ , a quasilocal measure, which tends towards the ADM measure as  $r \rightarrow \infty$ . For sufficiently isolated systems the quasilocal measure should well approximate the ADM measure at any radius greater than the systems. This detail is demonstrated in the context of Schwarzschild spacetime [42].

# Chapter 4

## Copernican Spacetime

### 4.1 Homogeneity & Isotropy

Classical cosmology begins with the Copernican principle, the statement that on sufficiently large scales the universe is spatially homogeneous and isotropic. The idea is supported by observation of the Cosmic Microwave Background and galactic structure and the premise that our position in the universe is not unique [4]. We impose spatial homogeneity and isotropy on the metric (3.4) and find  $N = N(t)$ ,  $N^i = 0$  and  $h_{ij}(t, x^i) = a^2(t)h_{ij}^0(x^i)$ ,

$$ds^2 = -N^2(t)dt^2 + a^2(t)h_{ij}^0(x^i)dx^i dx^j. \quad (4.1)$$

Eliminating the non-symmetric transformations under  $N^i$ , there remain seven degrees of freedom. The spatial metric  $h_{ij}^0$  must describe a maximally symmetric three-surface and the only possibilities are a three-sphere ( $\kappa = 1$ ), three-hyperboloid ( $\kappa = -1$ ) or three-plane ( $\kappa = 0$ ). On the three-sphere, we conveniently repack the spatial metric into

$$ds^2 = -N^2(t)dt^2 + a^2(t)d\Omega_3^2. \quad (4.2)$$

This is the closed FLRW metric, a cornerstone of modern classical cosmology. We insist that the universe is closed, spatially homogeneous and isotropic, and we find remaining degrees of freedom in the lapse,  $N(t)$ , and the expansion factor,  $a(t)$ . The freedom in  $N(t)$ , means the rate of flow of ‘time’ is variable and can change throughout the universes lifetime. The freedom in  $a(t)$  allows us to describe a universe wherein the distance between two points can grow or shrink as time increases, a dynamical universe. Our objective is to quantise this spacetime following the procedure laid out in Chapter 2.

### 4.2 The Hilbert Space of a Universe

The Quantum Mechanical state of a system,  $|\psi\rangle \in \mathbb{H}$ , encodes all possible configurations of the system and the probability for the system to be in each configuration. For an electron, the state  $|\psi_e\rangle$  encodes the probability of being in a position  $x^i$  (or momentum  $p_i$ ) at time  $t$  through its wavefunction, a position (or momentum) representation of the state,

$$\psi(x) = \langle x|\psi\rangle, \quad (4.3)$$

$$\mathcal{P}(x = x_0) = |\psi(x_0)|^2. \quad (4.4)$$

The Hilbert space of electron states supersedes the configuration space of trajectories for a ‘classical electron’ upon quantisation. In order to translate this formalism to cosmology we first consider which quantities uniquely define a universe, its configuration space, and the Hilbert

space we must derive from it.

The configuration space of a universe, historically called Superspace, must specify the universes three-geometry and matter content. A candidate for this Superspace is

$$\text{Riem}(\Sigma_t) = \{h_{ij}(x), \Phi(x) | x \in \Sigma_t\}, \quad (4.5)$$

except this identifies universes related by diffeomorphisms as unique. As only the three-geometry, not the three-metric, should uniquely assign a universe, we take the quotient [34]

$$\mathcal{S}(\Sigma_t) = \text{Riem}(\Sigma_t) / \text{Diff}(\Sigma_t). \quad (4.6)$$

As the Momentum constraints generate diffeomorphism invariance on the three-surface, imposing the constraints is equivalent to working in Superspace. We expect universe states to be elements of the Hilbert space derived from our infinite dimensional Superspace,  $|\Psi\rangle \in \mathbb{H}[\mathcal{S}(\Sigma_t)]$ . When we impose the Copernican principle and eliminate the shift, we reduce the configuration space of allowed universes from Superspace to a finite dimensional Minisuperspace. We further restrict the Minisuperspace by only looking at closed geometries,  $\kappa = 1$ . The Hilbert space of this Minisuperspace is the arena of our Quantum Cosmology.

### 4.3 General Behaviour of Wheeler-DeWitt

We quantise the gravitational Hamiltonian (3.18) in Minisuperspace in the metric representation such that it acts on the wavefunctional  $\Psi[h_{ij}(x)]$ ,

$$\hat{\mathcal{H}}\Psi[h_{ij}(x)] = N \left( -\hbar^2 G_{ijkl} \frac{\delta}{\delta h_{ij}} \frac{\delta}{\delta h_{kl}} - \sqrt{\hbar} ({}^{(3)}R - 2\Lambda) \right) \Psi[h_{ij}(x)] \approx 0. \quad (4.7)$$

This is the Wheeler-DeWitt equation on Minisuperspace, whose solution is a wavefunction of the universe. A unique solution requires sufficient boundary conditions.

Following [32], we argue for the general behaviour of Wheeler-DeWitt solutions in Minisuperspace. When dealing with specific systems, we can split the degrees of freedom into a timelike  $q^0$  and spacelike  $\mathbf{q}$ , such that under  $\hbar = 1$ , (4.7) generally reads as

$$\left( \frac{\partial^2}{\partial q_0^2} - \frac{\partial^2}{\partial \mathbf{q}^2} + U(q_0, \mathbf{q}) \right) \Psi = 0. \quad (4.8)$$

The behaviour of the solutions depends entirely on the behaviour of  $U(q_0, \mathbf{q})$ , particularly on whether its dependence is primarily timelike or spacelike. The argument we follow considers surfaces of constant  $U(q_0, \mathbf{q})$  in Minisuperspace. If they are all spacelike, then we can always perform a transformation such that the potential behaves entirely timelike,  $U = U(q_0)$ . In this way, the equation is separable with a separation constant,  $\lambda$ . For sufficiently large  $|U|$ , we can take the potential to dominate the separation constant. The solution is oscillatory for  $U \gg 0$  and exponential for  $U \ll 0$  and the converse is true for timelike surfaces of potential.

In the oscillatory regions, the potential is sufficiently ‘slowly varying’ that explicit solutions can be found via the WKB approximation [43], where the wavefunction is given by an ansatz defined in terms of a function  $S(q)$ :

$$\Psi(q) = C(q) \exp(-iS(q)). \quad (4.9)$$

If this wavefunction satisfies the Wheeler-DeWitt equation, then  $S(q)$  must satisfy the Lorentzian Hamilton-Jacobi equation in the classical limit  $\hbar \rightarrow 0$ :

$$\frac{1}{2}(\nabla S)^2 + U(q) = 0. \quad (4.10)$$

In this sense,  $S(q)$ , is clearly related to the classical action. Outside of the classical limit, we expand  $S(q) = S_0(q) + \hbar S_1(q) + \mathcal{O}(\hbar^2)$ . Ignoring higher order terms we substitute this into (4.10) and solve for  $S_0$  and  $S_1$  which are used to explicitly write out the approximate wavefunction.

Each choice of separation constant gives a different  $\lambda$ -mode of the wavefunction, analogous to a frequency of vibration. The true wavefunction should then be a superposition over these  $\lambda$ -modes with respect to some amplitude function,  $\mathcal{A}(\lambda)$ , which vanishes as  $|\lambda| \rightarrow \infty$  to ensure normalisation. The amplitude function allows us to construct physical wavepackets,

$$\Psi(q) = \int_{\mathbb{R}} d\lambda \mathcal{A}(\lambda) \Psi_{\lambda}(q). \quad (4.11)$$

Due to the highly oscillatory nature of the wavefunction, a significant contribution to the wavepacket is only made when the phase is stationary relative to  $\lambda$ ,

$$\frac{\partial S_{\lambda}}{\partial \lambda} = 0. \quad (4.12)$$

This produces the classical equation of motion and means the peak (ridge) of the wavefunction will always follow the classical trajectory in configuration space. The importance of this cannot be understated; there must exist a region where the quantum theory corresponds with the classical theory and in the WKB approximation there is. The approximation breaks down at the turning point of  $U(q)$  where  $p_i = 0$ .

## 4.4 Hamiltonian Cosmology

Applying our machinery to the metric (4.1), the Einstein-Hilbert action (3.10) reduces to

$$S_{EH} = \frac{1}{2} \int_{\mathbb{R}} dt N \left( -\frac{a\dot{a}}{N^2} + a - \frac{\Lambda a^3}{3} \right). \quad (4.13)$$

As the three-sphere is boundaryless, the surface terms cancel exactly. The matter content of this universe will be a single scalar field,  $\varphi(x)$ . Although in principle we can produce a generic matter content,  $\Phi(x)$ , from a number of fields, the inclusion of fields with spin is problematic [44]. After scaling, we write the Klein-Gordon action as

$$\begin{aligned} S_M &= \int_{\mathcal{M}} d^4x \sqrt{-g} \left( -\frac{1}{2} g^{\mu\nu} \varphi_{,\mu} \varphi_{,\nu} - m^2 \varphi^2 \right) \\ &= \frac{1}{2} \int_{\mathbb{R}} dt N a^3 \left( \frac{\dot{\varphi}^2}{N^2} - m^2 \varphi^2 \right). \end{aligned} \quad (4.14)$$

The total action,  $S_T = S_{EH} + S_M$ , can be written in terms of a Minisuperspace DeWitt metric where indices are grouped  $(ij) = A$ ,  $(kl) = B$  and generalised co-ordinates  $q^A \in \{a, \varphi\}$  [34],

$$S_T = \int_{\mathbb{R}} dt N \left( \frac{1}{2N^2} G_{AB} \dot{q}^A \dot{q}^B - V(q) \right), \quad (4.15)$$

$$G_{AB} = \begin{pmatrix} -a & 0 \\ 0 & a^3 \end{pmatrix}, \quad (4.16)$$

$$V(q) = \frac{1}{2} \left( -a + a^3 \left( \frac{\Lambda}{3} + m^2 \varphi^2 \right) \right). \quad (4.17)$$

Variation of this action reproduces the classical Friedmann equations of motion which determine the dynamics of the expansion factor, lapse and scalar field. Post-fixing  $N = 1$  the equations reduce to

$$\dot{a}^2 = -1 + a^2 \left( \dot{\varphi}^2 + \frac{\Lambda}{3} + m^2 \phi^2 \right), \quad (4.18)$$

$$\ddot{\varphi} + \frac{3\dot{a}}{a} \dot{\varphi} + m^2 \varphi^2 = 0. \quad (4.19)$$

Under  $m = 0$ , (4.19) tells us that  $\bar{\lambda} = a^3 \dot{\varphi}$  is a classical constant of motion. The equations of motion can be expressed parametrically and solved analytically to produce

$$\varphi(a) = \pm \frac{1}{2} \operatorname{arccosh} \left( \frac{\bar{\lambda}}{a^2} \right) + K. \quad (4.20)$$

The scalar,  $\varphi$ , acts as an inflaton field. Its coupling to the expansion factor,  $a$ , drives growth in the early universe when  $a$  is small. This growth slows once the universe is of sufficient size. We impose  $K = 0$ , to ensure a symmetric shrinking occurs in the universe near the end of its lifetime. The inflationary era in this universe resolves the horizon problem.

Transitioning to the Hamiltonian, we calculate the conjugate momenta using (2.1), finding that

$$p_N \approx 0, \quad (4.21)$$

$$p_a = -\frac{a\dot{a}}{N}, \quad (4.22)$$

$$p_\varphi = \frac{\bar{\lambda}}{N}. \quad (4.23)$$

The vanishing momentum (4.21) produces a Hamiltonian constraint  $\mathcal{H} \approx 0$  which we see within the naive Hamiltonian from the Legendre transform,

$$\mathcal{H} = N \left( \frac{1}{2N^2} G^{AB} p_A p_B + V(q) \right) \approx 0, \quad (4.24)$$

As mentioned earlier, there are ambiguities in operator ordering upon quantisation. In this context, where the momenta are grouped by a DeWitt metric, we resolve this by replacing the kinetic term with a Laplace-Beltrami operator [45],

$$G^{AB} p_A p_B \rightarrow -\hbar^2 \nabla \equiv -\hbar^2 \frac{1}{\sqrt{-G}} \partial_A (\sqrt{-G} G^{AB} \partial_B). \quad (4.25)$$

This replacement ensures the Hamiltonian is covariant on the Superspace. Post-fixing  $N = 1$  again, the Wheeler-DeWitt equation derived from our Hamiltonian reads

$$\frac{1}{2} \left( \frac{\hbar^2}{a^2} \frac{\partial}{\partial a} \left( a \frac{\partial}{\partial a} \right) - \frac{\hbar^2}{a^3} \frac{\partial^2}{\partial \varphi^2} + 2V(a, \varphi) \right) \Psi(a, \varphi) = 0. \quad (4.26)$$

## 4.5 Minisuperspace Wavepackets

Conventionally, at this point one begins to discuss boundary proposals [12, 11, 32]. Although they answer some important questions, our focus is to develop the machinery and interpretation of Minisuperspace wavepackets. To that end, we review work on a specific model: A closed

cosmology with zero cosmological constant, permeated by a massless scalar field [33]. Under  $m = 0$ ,  $\Lambda = 0$ ,  $\hbar = 1$  and  $\alpha = \ln a$ , the Wheeler-DeWitt equation (4.26) reduces to

$$\left( \frac{\partial^2}{\partial \alpha^2} - \frac{\partial^2}{\partial \varphi^2} - \exp(4\alpha) \right) \Psi = 0. \quad (4.27)$$

Assuming a separable solution of the form  $\Psi(\alpha, \varphi) = A(\alpha)\Phi(\varphi)$ , we split the partial differential equation into two families of ordinary differential equations:

$$\frac{d^2 A_\lambda}{d\alpha^2} - (\exp(4\alpha) - \lambda^2)A_\lambda = 0, \quad (4.28)$$

$$\frac{d^2 \Phi_\lambda}{d\varphi^2} + \lambda^2 \Phi_\lambda = 0, \quad (4.29)$$

where  $\lambda$  is a separation constant. Once again we change variables,  $\mathbf{a} = \frac{1}{2} \exp(2\alpha) = \frac{1}{2} a^2$ , to write (4.28) in the form of a hyperbolic (modified) Bessel equation [46],

$$\mathbf{a}^2 \frac{d^2 A_\lambda}{d\mathbf{a}^2} + \mathbf{a} \frac{dA_\lambda}{d\mathbf{a}} - \left[ \mathbf{a}^2 + \left( \frac{i\lambda}{2} \right)^2 \right] A_\lambda = 0, \quad (4.30)$$

which has a solution in terms of the hyperbolic Bessel functions and constants  $c_I$  and  $c_K$ ,

$$A_\lambda = c_I I_{\frac{i\lambda}{2}}(\mathbf{a}) + c_K K_{\frac{i\lambda}{2}}(\mathbf{a}). \quad (4.31)$$

We recognise that  $I_{\frac{i\lambda}{2}}(\mathbf{a})$  diverges for large  $\mathbf{a}$  such that we discard it to ensure a normalisable wavepacket. Looking at (4.29) we see a familiar friend, the harmonic oscillator,

$$\Phi_\lambda = c_+ \exp(i\lambda\varphi) + c_- \exp(-i\lambda\varphi), \quad (4.32)$$

with constants  $c_+$  and  $c_-$  to be fixed by boundary conditions. In this solution the scalar degree of freedom,  $\varphi$ , can act as a physical clock, an intrinsic measure of time in a system with no extrinsic one [36]. The conjugate variable,  $\lambda$ , is the frequency, measuring how fast ‘clock time’ runs relative to co-ordinate time. We do not take this idea further here, but will return to it when we study Schwarzschild spacetime.

Although we have an exact solution (4.31) of (4.28), we WKB approximate in the oscillatory region,  $\alpha < \bar{\alpha}$ , in order to discard higher order terms which would otherwise confuse our interpretation,

$$A_\lambda(\alpha) = (\lambda^2 - \exp(4\alpha))^{-\frac{1}{4}} \cos \left( \frac{\lambda}{2} \operatorname{arccosh} \left( \frac{\lambda}{\exp(2\alpha)} \right) - \frac{1}{2} \sqrt{\lambda^2 - \exp(4\alpha)} - \frac{\pi}{4} \right). \quad (4.33)$$

Treating either the positive or negative frequency solution in  $\Phi_\lambda$ , we combine with  $A_\lambda$ , and express the  $\lambda$ -mode wavefunction in WKB form (4.9) with phase

$$S_\lambda(\alpha, \varphi) = \frac{\lambda}{2} \operatorname{arccosh} \left( \frac{\lambda}{\exp(2\alpha)} \right) - \frac{1}{2} \sqrt{\lambda^2 - \exp(4\alpha)} - \frac{\pi}{4} \pm \lambda\varphi. \quad (4.34)$$

As mentioned, the wavefunction only sees a significant contribution where the phase is stationary with respect to  $\lambda$  such that (4.12) is satisfied. In fact, the condition exactly reproduces the classical equation of motion,

$$\varphi(a) = \pm \frac{1}{2} \operatorname{arccosh} \left( \frac{\lambda}{a^2} \right). \quad (4.35)$$

This is crucial, we have found correspondence between the classical and quantum theories. To fully align them, we centre the free parameter in the quantum theory  $\lambda$ , around a choice of classical constant of motion  $\bar{\lambda}$ , by taking the amplitude function to be a Gaussian of width  $b$  centred on  $\bar{\lambda}$  (non-standard normalisation) [34],

$$\mathcal{A}(\lambda) = \frac{1}{\sqrt{b\sqrt{\pi}}} \exp\left(-\frac{(\lambda - \bar{\lambda})^2}{2b^2}\right). \quad (4.36)$$

The ridge of the wavepacket evaluated by integrating over this amplitude function should follow the classical trajectory (4.35) corresponding to a classical constant of motion within this Gaussian range,

$$\Psi(\alpha, \varphi) = \int_{\mathbb{R}} d\lambda \mathcal{A}(\lambda) \Psi_{\lambda}(\alpha, \varphi). \quad (4.37)$$

Absorbing constants,  $c_i$ , into the amplitude we perform the integral, approximating  $\lambda$  terms ‘slowly varying’ with respect to the width of the Gaussian as constants evaluated on  $\bar{\lambda}$ ,

$$\begin{aligned} \Psi = \frac{\pi^{\frac{1}{4}}}{2\sqrt{b}} \frac{\exp(-\bar{\lambda}\pi)}{(\bar{\lambda}^2 - \exp(4\alpha))^{\frac{1}{4}}} \int_{\mathbb{R}} d\lambda \exp\left(\frac{(\lambda - \bar{\lambda})^2}{2b^2}\right) \\ \exp(-i\lambda\varphi) \cos\left(\frac{\lambda}{2} \operatorname{arccosh}\left(\frac{\bar{\lambda}}{\exp(2\alpha)}\right) - \frac{1}{2}\sqrt{\lambda^2 - \exp(4\alpha)} - \frac{\pi}{4}\right). \end{aligned} \quad (4.38)$$

Analysing  $\Re(\Psi)$ , we finally see the physics in two Gaussian wavepackets of width  $b^{-1}$ . Both wavepacket ridges follow the classical trajectory (4.35) for  $\bar{\lambda}$ , one follows the positive branch while the other follows the negative. The superposition tells us that the state most probably follows the classical trajectory (either positive or negative) for the constant of motion  $\bar{\lambda}$ , but that deviations from both the trajectory and the constant of motion are possible due to quantum uncertainty.

# Chapter 5

## Schwarzschild Spacetime

### 5.1 Spherical Symmetry & Staticity

Spherically symmetric spacetime admits an  $SO(3)$  group of isometries. A generic spherically symmetric metric is foliated in the orbits of this group: two-surfaces of constant positive curvature and two-surfaces of indefinite curvature. Without loss of generality, we foliate the manifold in leaves of  $S^2$  parameterised by  $\{\theta, \phi\}$  and orthogonal two-surfaces parameterised by  $\{t, r\}$  such that the metric reads [4]

$$ds^2 = d\tau^2(t, r) + S^2(t, r)d\Omega_2^2(\theta, \phi). \quad (5.1)$$

We are free to define  $t$  and  $r$  to be orthogonal on the two-surface and remain arbitrary such that we can expand  $d\tau^2$  in  $dt^2$  and  $dr^2$ ,

$$ds^2 = -T(t, r)dt^2 + \frac{dr^2}{R(t, r)} + S^2(t, r)d\Omega_2^2. \quad (5.2)$$

To explicitly see that we can impose orthogonality, imagine that  $t$  and  $r$  were not orthogonal, such that there existed  $dt dr$  cross terms in the metric. We could arbitrarily redefine  $t$  to eliminate the cross terms while still keeping a free function [47]. By imposing spherical symmetry, we give ourselves the freedom to eliminate the non-symmetric physical transformations described by the shift  $N^i$ .

Static spacetime admits a timelike Killing vector,  $\xi^\mu$ , which is hypersurface orthogonal such that it satisfies the condition,  $\xi_{[\mu}\nabla_\rho\xi_{\nu]} = 0$ . While the presence of the Killing vector implies that the metric components cannot be functions of the timelike co-ordinate,  $t$ , spacelike hypersurface orthogonality eliminates all  $dt dx^i$  cross terms. A generic static metric therefore takes the form [48]

$$ds^2 = -V^2(x^i)dt^2 + h_{ij}(x^i)dx^i dx^j. \quad (5.3)$$

The Schwarzschild solution is a spherically symmetric and static spacetime which satisfies the vacuum Einstein equations. Under a gauge choice (to be discussed later), the ansatz (5.2) produces the classical solution

$$ds^2 = -\left(1 - \frac{2m}{r}\right)dt^2 + \frac{dr^2}{1 - \frac{2m}{r}} + r^2 d\Omega_2^2. \quad (5.4)$$

The boundary  $r = 2m$  behaves as an event horizon, separating causally disconnected regions of spacetime, such that the metric describes a black hole. Strictly speaking, staticity is not necessary to produce Schwarzschild spacetime as the though the metric is static, it is derived only under the assumption of spherical symmetry. There is an overlap between spherical symmetry and staticity which is formally understood through [49, 50]:

**Birkhoff's Theorem.** *Any spherically symmetric  $C^2$  solution of the vacuum Einstein equations is locally equivalent to Schwarzschild and is thus static.*

To study fluctuations of the Schwarzschild background, we look for a metric ansatz which is generically spherically symmetric such that on-shell the solution is Schwarzschild. While imposing staticity may be seen as an over-fixing, we find this acceptable as we intend to study structure rather than dynamics. Our starting point is then

$$ds^2 = -T(r)dt^2 + \frac{dr^2}{R(r)} + S^2(r)d\Omega_2^2. \quad (5.5)$$

## 5.2 Classical Formulation

Once again, our objective is to generate an expression of the physics in Lagrangian and then Hamiltonian form. We find a gravitational Lagrangian from the metric (5.5) using the Einstein-Hilbert action, but impose a slight variation. We mathematically trade  $t$ -evolution for  $r$ -evolution such that  $\mathcal{L}(q, q', r)$  is analogous to  $\mathcal{L}(q, \dot{q}, t)$  [20]. As a consequence, the (geodesic) foliation is in  $r$  and we can integrate by parts where appropriate to eliminate second order terms. Ignoring the Gibbons-Hawking-York term,

$$\begin{aligned} S_{EH} &= \int_{\mathcal{M}} d^4x \sqrt{-g} \mathbf{R} \\ &= \int_{\mathbb{R}} dr L = \int_{\mathbb{R}} dr \int_{\Sigma_r} dt d\theta d\phi \mathcal{L}. \end{aligned} \quad (5.6)$$

We explicitly write out the terms in the integrand evaluated on the metric (5.5),

$$\sqrt{-g} = \sqrt{\frac{T}{R}} S^2 \sin \theta, \quad (5.7)$$

$$\mathbf{R} = -\frac{4RS''}{S} - \frac{2RS'^2}{S^2} - \frac{RT''}{T} - \frac{2RT'S'}{TS} + \frac{RT'^2}{2T^2} - \frac{2R'S'}{S} + \frac{2}{S^2} - \frac{R'T'}{2T}. \quad (5.8)$$

Integrating by parts to eliminate second derivatives (discarding boundary terms) and integrating over the solid angle (absorbing  $4\pi$ ) we produce the first order Lagrangian [23]

$$\mathcal{L} = -\sqrt{\frac{T}{R}} \left( 1 + RS'^2 + \frac{RT'SS'}{T} \right), \quad (5.9)$$

for generalised co-ordinates  $q^i \in \{T, R, S\}$ . Note that this Lagrangian has no explicit  $r$  dependence. As this formulation is equivalent to Einstein's equations, we expect the solution of the Euler-Lagrange equations to be Schwarzschild. Generating these equations of motion for generalised co-ordinates  $T$ ,  $R$  and  $S$ ,

$$\sqrt{\frac{R}{T}}(-1 + RS'^2 + S(R'S' + 2RS'')) = 0, \quad (5.10)$$

$$\frac{1}{\sqrt{RT}}(T(-1 + RS'^2) + RSS'T') = 0, \quad (5.11)$$

$$\frac{1}{T}\sqrt{\frac{R}{T}}(-RST'^2 + 2T^2(R'S' + 2RS'') + T(2RS'T' + S(R'T' + 2RT''))) = 0. \quad (5.12)$$

This system of equations does not have an analytic solution, but post-fixing  $S = r$  (see [Appendix 2](#) for the role of  $S$ ), the  $T$  (5.10) and  $R$  (5.11) equations of motion reduce to

$$\frac{R'}{R} + \frac{1}{r} - \frac{1}{Rr} = 0, \quad (5.13)$$

$$1 + \frac{rT'}{T} - \frac{1}{R} = 0. \quad (5.14)$$

These equations can be solved exactly to reproduce the classical solution up to a choice of integration constants,  $R = 1 + \frac{c_1}{r}$  and  $T = c_2 R$ . The  $S$  (5.12) equation trivially reduces to  $0 = 0$  upon substitution of the  $T$  and  $R$  solutions.

We define the conjugate momenta according to (2.1) but as we traded evolution between  $t$  and  $r$ , the derivative is taken with respect to  $q'^i$  rather than  $\dot{q}^i$ ;

$$p_T = -SS' \sqrt{\frac{R}{T}}, \quad (5.15)$$

$$p_R \approx 0, \quad (5.16)$$

$$p_S = -\sqrt{\frac{T}{R}} \left( 2RS' + \frac{RST'}{T} \right). \quad (5.17)$$

To pass over to the Hamiltonian, we Legendre transform the Lagrangian (5.9), producing

$$\mathcal{H} = \sqrt{\frac{T}{R}} \left( 1 + \frac{Tp_T^2}{S^2} - \frac{p_T p_S}{S} \right). \quad (5.18)$$

Drawing analogy with methods in Quantum Cosmology, we relabel  $\frac{1}{R} \rightarrow N^2$  to clearly denote the lapse function (acting on  $dr^2$  rather than  $dt^2$  due to the traded evolution)

$$\mathcal{H} = N\sqrt{T} \left( 1 + \frac{Tp_T^2}{S^2} - \frac{p_T p_S}{S} \right). \quad (5.19)$$

The lapse,  $N$ , viewed either as a Lagrange multiplier or through the constraint  $p_N \approx 0$ , generates a Hamiltonian constraint such that the Hamiltonian (5.19) must vanish on-shell:

$$\mathcal{H} = \sqrt{T} \left( 1 + \frac{Tp_T^2}{S^2} - \frac{p_T p_S}{S} \right) \approx 0. \quad (5.20)$$

### 5.3 Wormhole Co-ordinates

Direct attempts to quantise the Hamiltonian (5.19) suffer ambiguities in the cross momentum and inverse terms. We circumvent these problems by producing an equivalent classical theory without these defects (see [Appendix 2](#) and [Appendix 3](#) for alternate quantisations). To eliminate the cross momentum term, we complete the square, treating the Hamiltonian as a quadratic in  $p_T$ :

$$\mathcal{H} = N\sqrt{T} \left[ \frac{T}{S^2} \left( p_T - \frac{Sp_S}{2T} \right)^2 - \frac{p_S^2}{4T} + 1 \right]. \quad (5.21)$$

This restructuring makes obvious a potential canonical transformation which will allow us to trade complexity in the Hamiltonian for complexity in the canonical variables. Retaining  $N$

from the original co-ordinates, we are free to choose positive square roots for our new co-ordinates,

$$X = \ln T, \quad (5.22)$$

$$Y = 2 \ln S + \ln T, \quad (5.23)$$

$$p_X = T p_T - \frac{S p_S}{2}, \quad (5.24)$$

$$p_Y = \frac{S p_S}{2}. \quad (5.25)$$

To verify that the transformation is canonical and the symplectic structure is preserved we calculate  $\{X, Y\} = \{X, N\} = \{Y, N\} = 0$ ,  $\{p_X, p_Y\} = \{p_X, p_N\} = \{p_Y, p_N\} = 0$ ,  $\{X, p_Y\} = \{X, p_N\} = \{Y, p_X\} = \{Y, p_N\} = \{N, p_X\} = \{N, p_Y\} = 0$  and  $\{X, p_X\} = \{Y, p_Y\} = \{N, p_N\} = 1$  [51]. The transformed classical Hamiltonian is then given by

$$\mathcal{H} = N \exp\left(\frac{X}{2}\right) [\exp(-Y)(p_X^2 - p_Y^2) + 1], \quad (5.26)$$

and reparameterising the metric in terms of these transformed co-ordinates we see that

$$ds^2 = N^2 dr^2 - \exp(X) dt^2 + \exp(Y - X) d\Omega_2^2. \quad (5.27)$$

In this form, Hamilton's equations of motion are made exceptionally simple via the Poisson bracket. Along with  $\mathcal{H} \approx 0$  we expect

$$X' = N \exp\left(\frac{X}{2} - Y\right) 2p_X, \quad (5.28)$$

$$Y' = -N \exp\left(\frac{X}{2} - Y\right) 2p_Y, \quad (5.29)$$

$$p'_X = 0, \quad (5.30)$$

$$p'_Y = -N \exp\left(\frac{X}{2}\right). \quad (5.31)$$

As in the cosmological case, we clearly see that the lapse,  $N$ , describes the gauge degree of freedom in this system. We post-fix  $N = \exp\left(-\frac{X}{2}\right)$ , which in the old co-ordinates is equivalent to fixing  $T = R$ . The momentum equations of motion, (5.30) and (5.31), are then solved by

$$p_X = m, \quad (5.32)$$

$$p_Y = -r + c, \quad (5.33)$$

where  $m$  and  $c$  are both constants of integration. Note that post-fixing is not necessary to obtain the  $p_X$  solution such that it is the same constant of motion in any gauge (or no gauge). The Hamiltonian constrains these two constants to be related as  $c = \pm m$ , leading to two solutions for (5.28) and (5.29), a positive and negative branch,

$$\exp(X_{\pm}) = \left(1 \mp \frac{2m}{r}\right)^{\pm 1}, \quad (5.34)$$

$$\exp(Y_+ - X_+) = r^2, \quad (5.35)$$

$$\exp(Y_- - X_-) = r^2 \left(1 + \frac{2m}{r}\right)^2. \quad (5.36)$$

In solving for these we eliminate a constant of integration through a shift in the co-ordinates. The positive branch exactly reproduces the Schwarzschild solution (5.4) such that it should be clear why we labelled the  $p_X$  integration constant  $m$ : The only gauge invariant constant of motion we can expect in a Schwarzschild spacetime is the mass. The negative branch is more subtle but under the co-ordinate transformation  $r \rightarrow r - 2m$  it also reproduces Schwarzschild exactly. Parametrically, the classical trajectory through configuration space (identical for both branches such that we write  $X = X_{\pm}$ ) is

$$\sinh\left(\frac{X}{2}\right) = -m \exp\left(-\frac{Y}{2}\right). \quad (5.37)$$

This classical trajectory, see Figure 5.1, covers  $X < 0$ ,  $r > 0$ .

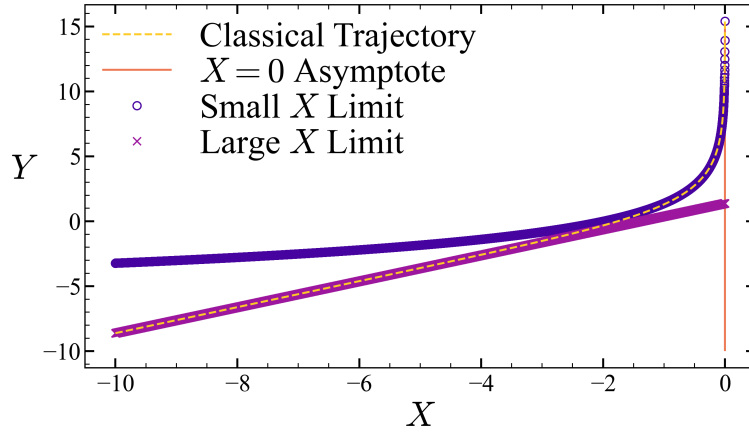


Figure 5.1: Configuration space,  $X - Y$ , plot of the  $m = 1$  classical trajectory on the range  $X \in [-10, 0.0001]$ . The curve is asymptotic at  $X = 0$ . For  $X > -1$  the behaviour is logarithmic. For  $X < -1$ , the behaviour is linear.

In the asymptotic limit  $X \gg -1$ , we Taylor expand, and in the large limit  $X \ll -1$ , the negative exponential in  $\sinh$  dominates such that in these regions we write

$$Y = 2 \log\left(-\frac{2m}{X}\right), \quad X \gg -1, \quad (5.38)$$

$$Y = X + 2 \log(2m), \quad X \ll -1. \quad (5.39)$$

Though the classical trajectory exists only on  $X < 0$ , the map between  $r$  and  $X_{\pm}$  still exists outside of that range as seen in Figure 5.2.

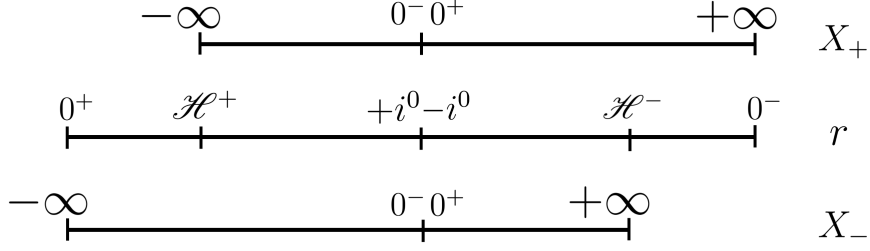


Figure 5.2: In the classical limit, we map ranges of  $r$  to ranges of  $X_{\pm}$  in terms of  $\mathcal{H}^{\pm} = \pm 2m$  and  $\pm i^0 = \pm \infty$ .  $X_+$  covers  $r \in (-i^0, 0^-) \cup (\mathcal{H}^+, +i^0)$ , excluding  $[0^+, \mathcal{H}^+]$ .  $X_-$  covers  $r \in (-i^0, \mathcal{H}^-) \cup (0^+, +i^0)$ , excluding  $[\mathcal{H}^-, 0^-]$ .

Curiously we describe a region of negative radius and although that seems absurd,  $r$  is simply a co-ordinate such that we can analytically continue into the range  $-\infty < r < +\infty$ . To cover this range, we must use patches of  $X_+$  and  $X_-$  as neither can capture  $r$  in its entirety. Under the transformation  $X_- \rightarrow -X_-$ , the negative branch metric can be written as

$$ds_-^2 = - \left( 1 + \frac{2m}{r} \right) dt^2 + \frac{dr^2}{1 + \frac{2m}{r}} + r^2 d\Omega_2^2, \quad (5.40)$$

which we interpret as Schwarzschild under  $r \rightarrow -r$  or  $m \rightarrow -m$ .

Putting everything together, the positive branch describes positive radius Schwarzschild spacetime and the negative branch (under a co-ordinate transformation) describes negative radius Schwarzschild spacetime such that we see two distinct but isometric regions of spacetime. Note that restricting the positive branch to positive radius and the negative branch to negative radius means that we cannot describe the regions  $[0^+, \mathcal{H}^+]$  and  $[\mathcal{H}^-, 0^-]$ . What we have done by defining new co-ordinates and analytically continuing into regions not previously accessible should be familiar from the Kruskal-Szekeres parameterisation [48]. They describe a region of Schwarzschild spacetime, a black hole, a white hole, and an isometric region of Schwarzschild spacetime on the other side of an Einstein-Rosen bridge. While we cannot access black hole and white hole regions, we still see a wormhole because unlike Kruskal-Szekeres spacetime ours lies at asymptotic infinity rather than the singularity. Drawing the Penrose diagram for the spacetime in Figure 5.3, we see the result of our traded foliation.

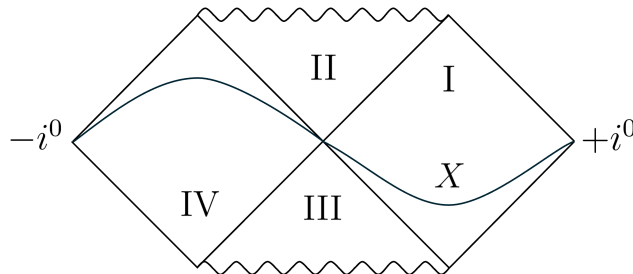


Figure 5.3: Penrose diagram divided into four regions I, II, III, and IV. Regions II and III are the black hole and white hole interiors respectively, neither are covered by our co-ordinates. Region I describes positive radius Schwarzschild spacetime and region IV describes negative radius Schwarzschild spacetime (isomorphic to positive radius Schwarzschild spacetime). Positive asymptotic infinity,  $+i^0$ , lies in region I. Negative asymptotic infinity,  $-i^0$ , lies in region IV. A trajectory of  $X$  is drawn between  $+i^0$  and  $-i^0$ .

What would conventionally be a line of constant  $t$  is now a trajectory following  $X$  between negative asymptotic infinity,  $-i^0$ , and positive asymptotic infinity,  $+i^0$ , similar to the hypersurface described in [22]. Travelling along  $X$  in region I through  $+i^0$ , we emerge from  $-i^0$  in region IV having passed through the wormhole. We do not claim that this wormhole is traversable, though our foliation is traded such that this trajectory is technically ‘timelike’ a real particle could not travel along it let alone reach infinity.

## 5.4 Quantisation

One peculiarity of quantisation is that equivalent classical theories do not produce equivalent quantum theories. This manifests most clearly in issues of operator ordering. As such, once we quantise in the new co-ordinates, we cannot go back to the old co-ordinates to probe the black hole interior. Under the usual quantisation, the Hamiltonian (5.26) becomes

$$\hat{\mathcal{H}} = \hat{N} \exp\left(\frac{\hat{X}}{2}\right) \left[ -\hbar^2 \exp(-\hat{Y}) \left( \frac{\partial^2}{\partial X^2} - \frac{\partial^2}{\partial Y^2} \right) + 1 \right]. \quad (5.41)$$

As  $\hat{N} \exp\left(\frac{\hat{X}}{2}\right)$  is a global pre-factor, we safely discard it to obtain a two variable phase space. Under  $\hbar = 1$  the Wheeler-DeWitt equation,  $\hat{\mathcal{H}}\Psi = 0$ , reduces to

$$\left( \frac{\partial^2}{\partial X^2} - \frac{\partial^2}{\partial Y^2} - \exp(Y) \right) \Psi = 0. \quad (5.42)$$

We recognise here the generic form of Wheeler-DeWitt equations (4.8), just as we saw with our closed cosmology (4.27). In the same spirit, we take a separable ansatz,  $\Psi(X, Y) = \chi(X)\gamma(Y)$ , with separation constant  $\lambda$ . The  $\chi_\lambda(X)$  solution is identical to  $\Phi_\lambda(\varphi)$ , a harmonic oscillator,

$$\chi_\lambda = c_+ \exp(i\lambda X) + c_- \exp(-i\lambda X), \quad (5.43)$$

again with constants  $c_+$  and  $c_-$  to be fixed by boundary conditions. In our cosmological discussion, we saw that a scalar field could act as a clock. In this system,  $X$  plays an analogous role to the scalar field,  $\varphi$ , such that we take  $X$  to be ‘time’. The frequency of the clock,  $\lambda$ , is then the conjugate momenta,  $p_X = m$ . With this interpretation, a superposition in  $\lambda$ -modes is now a superposition over  $m$ -modes. This interpretation is supported by the No-Hair theorem [52] which implies that a superposition of different black hole states could only be a superposition of masses. It also allows us to incorporate mass without resorting to the ADM energy (3.26) (see Appendix 1). The two independent solutions of  $\chi_m$  represent forwards and reverse time (or positive and negative mass).

Dealing with the other half of the separation, the  $\gamma_m(Y)$  equation differs from (4.28) as

$$\frac{d^2 \gamma_m}{dY^2} + (\exp(Y) + m^2) \gamma_m = 0. \quad (5.44)$$

Under the change of variable,  $\mathcal{Y} = \exp(Y)$ , we generate the Bessel-like equation

$$\mathcal{Y}^2 \frac{d^2 \gamma_m}{d\mathcal{Y}^2} + \mathcal{Y} \frac{d\gamma_m}{d\mathcal{Y}} + (\mathcal{Y} + m^2) \gamma_m = 0, \quad (5.45)$$

and although an analytic solution exists, it is not as straightforward as  $A_\lambda(\mathfrak{a})$ ,

$$\gamma_m = k_+ \Gamma(1 + 2im) J_{2im}\left(2\sqrt{\mathcal{Y}}\right) + k_- \Gamma(1 - 2im) J_{-2im}\left(2\sqrt{\mathcal{Y}}\right). \quad (5.46)$$

The constants  $k_+$  and  $k_-$  are fixed by boundary conditions. We are now in a position to build coherent wavepacket states treating  $X$  and  $m$  as Fourier duals under a Heisenberg uncertainty principle,  $\Delta X \Delta m = \frac{\hbar}{2}$ . Taking a square root Gaussian amplitude function (the inner product should be Gaussian distributed not the wavefunction), the generic wavefunction is then a superposition of  $m$ -modes,

$$\Psi = \int_{\mathbb{R}} dm \frac{1}{(2\pi\sigma^2)^{\frac{1}{4}}} \exp\left(-\frac{(m-m_0)^2}{4\sigma^2}\right) [c_+ \exp(imX) + c_- \exp(-imX)] \left[ k_+ \Gamma(1+2im) J_{2im}\left(2 \exp\left(\frac{Y}{2}\right)\right) + k_- \Gamma(1-2im) J_{-2im}\left(2 \exp\left(\frac{Y}{2}\right)\right) \right]. \quad (5.47)$$

For  $m_0 = 0$ , the integral describes an inner product of even functions such that we could ‘unproject’ the symmetry (analogous to symmetric and antisymmetric tensors) and discard the negative frequency term for either the oscillator or Bessel, effectively imposing  $c_- = 0$  or  $k_- = 0$ .

## 5.5 Wavepacket Analysis

The following section describes work in progress and its results are subject to change.

Analogous to our cosmology, the wavepacket probability distribution  $|\Psi|^2$  should exhibit a ridge in configuration space. This ridge should follow the  $m_0$  classical trajectory (5.37) in a region of classical correspondence and then diverge where quantum behaviour dominates. Through preliminary numerical integration, we project the ridge into the  $X - Y$  plane as contours and overlay the classical trajectory (5.37) in Figure 5.4.

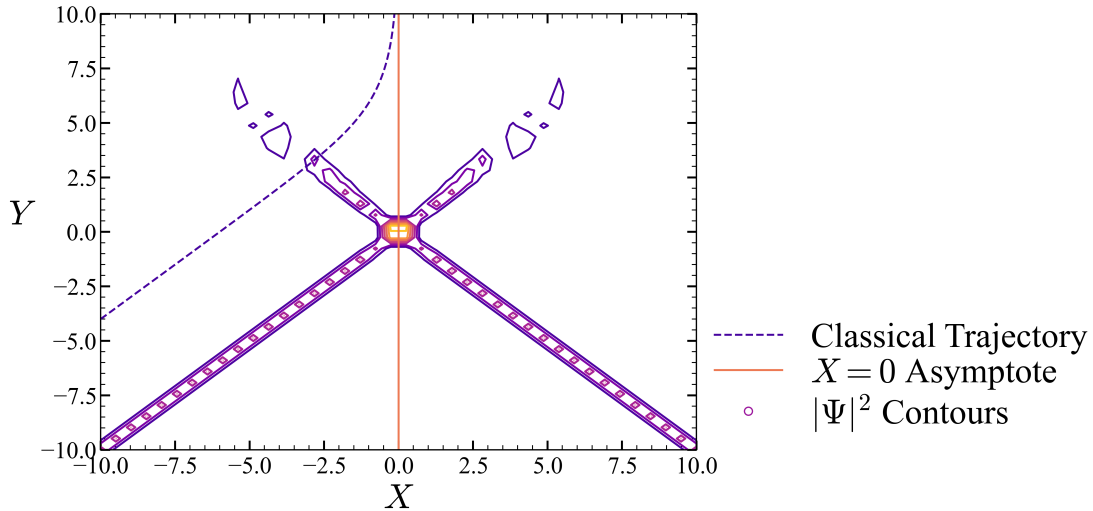


Figure 5.4: Configuration space,  $X - Y$ , plot of the  $m = 10$  classical trajectory and contours of  $|\Psi|^2$  on the range  $X \in [-10, 10]$ ,  $Y \in [-10, 10]$ . We fix  $c_+ = c_- = k_+ = k_- = 1$ ,  $m_0 = 10$  and  $\sigma = 1$  and integrate over the range  $m \in [-80, 100]$ . The contours form discrete peaks rather than a true ridge as an artefact of the number of points in our computation.

The numerical method fails to accurately capture the shift of the Gaussian to the correct mass,  $m_0$ , because the wavefunction diverges with respect to  $m$ . We expect the Gaussian to suppress these divergences but as they grow arbitrarily large, the computer calculation breaks

down. This is not to say that the integral does not converge, just that it cannot be numerically integrated in its current form.

Nevertheless we find that the quantum theory ‘agrees’ with the classical (for a correct shift) for  $X \ll -1$ , exhibiting linear behaviour of the form (5.39). This agreement breaks down for  $X \gg -1$  as the classical theory asymptotes to  $X = 0$ . These results are counter-intuitive, we see quantum behaviour at  $+i^0$  and classical behaviour at  $\mathcal{H}^+$ . Work is ongoing to understand this result but it is implied that quantum effects are significant near the wormhole.

Direct integration over the order of  $J_\nu(x)$  is avoided in the literature [17]. We too can avoid it by expressing the wavefunction in terms of hyperbolic Bessel functions which are relatively well-behaved. Through the relation [53]

$$K_\nu(z) = \frac{\pi}{2 \sin(\nu\pi)} (i^\nu J_{-\nu}(iz) - i^{-\nu} J_\nu(iz)), \quad (5.48)$$

we can fix  $k_+$  or  $k_-$ , to write  $\Psi$  in a more amenable form,

$$\begin{aligned} \Psi = \int_{\mathbb{R}} dm \frac{1}{(2\pi\sigma^2)^{\frac{1}{4}}} \exp\left(-\frac{(m-m_0)^2}{4\sigma^2}\right) \\ [c_+ \exp(imX) + c_- \exp(-imX)] \alpha(m) K_{2im}\left(\frac{2}{i} \exp\left(\frac{Y}{2}\right)\right). \end{aligned} \quad (5.49)$$

The function  $\alpha(m)$  is determined through analysis of boundary conditions and asymptotic limits. An appropriate set of boundary conditions could discard divergent behaviour within the integrand and allow us to further probe the physics in these coherent states.

# Chapter 6

## Conclusion

### 6.1 Overview

We conclude with an overview of our work on canonical quantisation of gravitational systems. Starting with [Chapter 1](#), we explained why a theory of Quantum Gravity is necessary and how quantisation of General Relativity could achieve that.

In [Chapter 2](#) we unraveled the complications which arise from constraints in Hamiltonian systems and clarified the relation between constraints and gauge degrees of freedom. We emphasised the importance of leaving these gauge degrees of freedom unfixed to avoid second-class constraints, which convolute the quantisation process. Finally, we described how to impose spacetime symmetries on a generic metric, a crucial process.

[Chapter 3](#) was a brief overview of canonical gravity where we covered the  $3 + 1$  decomposition of spacetime which allowed us to write a gravitational Lagrangian and Hamiltonian. Here we encountered the first major roadblock to a theory of Quantum Gravity: The Hamiltonian is constrained to vanish. We touched on the algebra of constraints and then finished by treating the boundary terms we had ignored in our earlier analysis. These boundary terms were applied to Schwarzschild spacetime in [Appendix 1](#).

We introduced our first gravitational system in [Chapter 4](#), a closed, homogeneous and isotropic cosmology permeated by a massless scalar field. Classically, the theory described a simple inflationary universe with a symmetric period of collapse near the end of its life. Quantum mechanically, using a WKB approximation, we built Gaussian wavepackets of the state which peaked on the classical trajectory of the system. The expectation trajectory was therefore the one which described classical inflation and collapse, but deviations from this were allowed due to quantum spread. Rather than study the system in great depth, we simply used it to build an understanding of wavepackets and their interpretation.

Finally, in [Chapter 5](#), we built a quantum theory of Schwarzschild spacetime. We began with a generic spherically symmetric and static spacetime which was Schwarzschild on-shell. The gravitational Hamiltonian we found from it was canonically transformed to a simple form in a set of ‘Wormhole’ co-ordinates. An analysis of these Wormhole co-ordinates showed us that the spacetime exhibited wormhole behaviour at asymptotic infinity. We then quantised the theory and performed a brief analysis of coherent wavepacket states and their classical correspondence although this analysis was left incomplete.

## 6.2 Outlook

There are many avenues still to be explored, most of them in the quantum theory. First and foremost, we must find an appropriate set of boundary conditions and impose them to solve for the function  $\alpha(m)$  and fix the other free parameters in the wavepacket. We should also look to formalise the quantum theory by establishing the Hilbert space in which the states exist in find an inner product. Only with an inner product can we verify the form of the probability distribution. Once the quantum theory is placed on formal foundations, we could make meaningful predictions about the wormhole and spacetime as a whole. We could make near horizon predictions and, if we reformulated with a different set of patches, probe the singularity.

# Appendix A

## Dead Ends

### A.1 The Inclusion of Boundary Terms

We treat the ADM mass (3.26) of the spherically symmetric and static spacetime (5.5) (ignoring surface terms from integration by parts). The first problem we face is that the mass is measured at a fixed distance such that to include mass in the Hamiltonian is to mix functions of  $r$  and those same functions evaluated at a given  $r_0$ . The second problem is that the ADM mass is measured at infinity, and as we expect quantum fluctuations to die off, the infinite limit will always be the on-shell limit. We cannot therefore expect to capture quantum behaviour in the ADM mass. We solve these problems by thinking about the Hamiltonian defined on the range  $r > 2m + \delta r$ , where  $\delta r$  is a scale of near-horizon fluctuations, and using a quasilocal mass measure. On the range  $2m + \delta r < r < \infty$ , any ADM-like observer should measure the mass as  $m$  (to a first approximation) such that we can take the boundary term to be a function of  $r$ , as whatever  $r_0$  we choose the answer is the same. Treating the foliation in the conventional way, in  $t$  rather than  $r$ , the surface energy integral for the spacetime metric (5.5) is

$$\oint_{S_\infty^2} d^2\sigma^i (h_{ij,j} - h_{jj,i}) = -\frac{40\pi}{3} S^3 S'. \quad (\text{A.1})$$

Eliminating  $S'$  using the momentum definition (5.15) and summing with the Hamiltonian (5.19) we obtain a Hamiltonian which accurately reflects the energy of the system such that it no longer vanishes,

$$\mathcal{H} = N\sqrt{T} \left( 1 + \frac{Tp_T^2}{S^2} - \frac{p_T p_S}{S} + cS^2 p_T \right). \quad (\text{A.2})$$

The constant  $c$ , is determined by absorbing constants consistently across the bulk and the boundary. We now perform a transformation to treat mass as a canonical variable such that we can meaningfully talk about superposition of mass,

$$M = cS^2 p_T. \quad (\text{A.3})$$

If a canonical transformation could be found, its conjugate variable could act as a physical clock in the spirit of unimodular gravity [16] and a Schrödinger equation could be obtained.

## A.2 Reduced Phase Space Quantisation

Looking at Hamilton's equations for the Hamiltonian (5.19) and generalised co-ordinates  $q^i \in \{T, N, S\}$ ,

$$T' = N \frac{\sqrt{T}}{S} \left( -p_S + 2 \frac{T}{S} p_T \right), \quad (\text{A.4})$$

$$N' = 0, \quad (\text{A.5})$$

$$S' = -N \sqrt{T} \frac{p_T}{S}, \quad (\text{A.6})$$

$$p_T' = -N \sqrt{T} \frac{p_T^2}{S^2} - \frac{\mathcal{H}}{2T} \approx N \sqrt{T} \frac{p_T^2}{S^2}, \quad (\text{A.7})$$

$$p_N' = -\sqrt{T} \left( 1 + \frac{T p_T^2}{S^2} - \frac{p_T p_S}{S} \right) = -\mathcal{H} \approx 0, \quad (\text{A.8})$$

$$p_S' = -N \sqrt{T} \left( \frac{p_T p_S}{S^2} - 2 \frac{T p_T^2}{S^3} \right). \quad (\text{A.9})$$

We note that the  $p_T$  (A.7) and  $S$  (A.6) equations of motion look similar in the on-shell limit. Combining these, we generate a differential equation which is uniquely solved by  $p_T = \alpha S$  for some constant  $\alpha$ . Substituting this result into the  $T$  (A.4) and  $p_S$  (A.9) equations of motion, we generate another differential equation which is solved by  $p_S = \alpha T + \beta$ . Substituting into the  $p_N$  (A.8) equation of motion (Hamiltonian constraint), we find that  $\beta = \frac{1}{\alpha}$  such that

$$p_T = \alpha S, \quad (\text{A.10})$$

$$p_S = \alpha T + \frac{1}{\alpha}. \quad (\text{A.11})$$

These relations can be further substituted into the  $S$  (A.6) and  $T$  (A.4) equations of motion and these can be combined to generate one more differential equation with solution

$$T = \frac{1}{\alpha^2} \left( 1 + \frac{c}{S} \right). \quad (\text{A.12})$$

For this to correspond with the classical limit, we take  $\alpha^2 = 1$  and  $c = -2m$  such that

$$T = 1 - \frac{2m}{S}. \quad (\text{A.13})$$

This relates two of our degrees of freedom, reducing the number of independent variables and the dimension of our phase space. As this is an on-shell relation and we are free to post-fix  $S = r$  to obtain Schwarzschild,  $S$  may be seen as a gauge degree of freedom but there is subtlety to this. Unlike  $N$  and  $N^i$  which are clearly gauge degrees of freedom,  $S$  is fixed through redefinition of the co-ordinate  $r$  such that fixing it cannot reduce the ten original degrees of freedom. As such  $S$  is not a true gauge degree of freedom of the metric although it remains a quantity that can be post-fixed in the equations of motion.

Another way to produce a quantisable Hamiltonian is to impose the relation (A.13) off-shell. Once we do so, we cannot expect  $p_T$  and  $p_S$  to be independent and applying the chain rule to the momentum definitions we find

$$p_T = p_S \frac{S^2}{2m}. \quad (\text{A.14})$$

With (A.13) and this momentum relation we can eliminate  $S$  in the Hamiltonian to obtain

$$\mathcal{H} = 1 + (T - 1)^2(2T - 1) \frac{p_T^2}{4m^2}. \quad (\text{A.15})$$

This classical Hamiltonian upon quantisation generates a Wheeler-deWitt equation,

$$\left( 1 - (T - 1)^2(2T - 1) \frac{\hbar^2}{4m^2} \frac{\partial^2}{\partial T^2} \right) \Psi \approx 0. \quad (\text{A.16})$$

The solution to this is hypergeometric, like many wavefunctions in Quantum Mechanics. Specifically, ours bears similarity to the Rosen-Morse potential [54],

$$\gamma = \frac{\sqrt{16m^2 + \hbar^2}}{\hbar}, \quad (\text{A.17})$$

$$\begin{aligned} \Psi(T) = & k_1 (T - 1)^{\frac{\gamma+1}{2}} {}_2F_1 \left( \frac{\gamma-1}{2}, \frac{\gamma+1}{2}; 0; 2T - 1 \right) \\ & + k_2 \left( T - \frac{1}{2} \right) (T - 1)^{\frac{\gamma+1}{2}} {}_2F_1 \left( \frac{\gamma+1}{2}, \frac{\gamma+3}{2}; 2; 2T - 1 \right). \end{aligned} \quad (\text{A.18})$$

Finding a correct interpretation is the tricky bit.

### A.3 Inverse Momentum Quantisation

The vanishing Hamiltonian (5.19) can be re-arranged for  $p_S$  as

$$p_S = \frac{S}{p_T} + \frac{T p_T}{S}. \quad (\text{A.19})$$

The inverse momentum operator quantised in the position representation is not well defined in the literature, but we can choose one which satisfies the expected properties of an inverse [55],

$$\begin{aligned} \phi(T) &= \hat{p}_T (\hat{p}_T^{-1} \phi(T)) \\ &= -i\hbar \frac{\partial}{\partial T} \left( \frac{i}{\hbar} \int_{-\infty}^T \phi(T') dT' \right). \end{aligned} \quad (\text{A.20})$$

Ambiguities at  $p_T = 0$  are avoided as the wavefunction decays sufficiently fast. Upon quantisation, we obtain a Schrödinger type equation with  $S$  as an intrinsic measure of time. This makes intuitive sense as the operator  $S$  is associated with the area of the black hole and thus with its entropy. As the arrow of time often follows from entropy it seems natural to define time in terms of entropy,

$$\begin{aligned} i\hbar \frac{\partial \psi}{\partial S} &= \mathcal{H}_{\text{effective}} \psi \\ &= i\hbar \frac{T}{S} \frac{\partial \psi}{\partial T} - \frac{iS}{\hbar} \int_{-\infty}^T \psi dT'. \end{aligned} \quad (\text{A.21})$$

This Hamiltonian still contains an inverse operator but the term may be eliminated under an appropriate approximation scheme.

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